SepsisDx Classification Report

Kaggle ICU Sepsis Dataset — 2,896 ICU Patients (Trained on Kaggle Data)
Generated March 06, 2026 at 15:43 | Model: Random Forest (threshold=0.3323) | Trained on 2,896 patients

Source Data & Training

Kaggle ICU Sepsis Dataset (Yilmaz Ba■er et al. 2025, *Biomedicines* 13(6):1449) — 2,896 ICU patients with CRP and Creatinine. Sepsis-3 criteria. 5-fold CV AUROC=0.941, threshold via Youden's J.

Classification Summary

Group	N	Correct	Accuracy	Avg P(sep)	LOW	MOD	HIGH	V.HIGH
Sepsis	703	660/703	93.9%	0.840	6	86	108	503
SIRS	259	238/259	91.9%	0.132	193	52	14	0
Healthy	1934	1738/1934	89.9%	0.128	1469	354	111	0
Overall	2896	2636/2896	91.0%	—	—	—	—	—

Sensitivity: 0.939 **Specificity:** 0.901 **PPV:** 0.753 **NPV:** 0.979 **TP:** 660 **TN:** 1976 **FP:** 217 **FN:** 43

Performance: Before vs After Kaggle Training

Metric	Before (Synthetic)	After (Kaggle)	Change
Accuracy	60.8%	91.0%	**+30.2%**
Sensitivity	87.2%	93.9%	**+6.7%**
Specificity	52.3%	90.1%	**+37.8%**
PPV	36.9%	75.3%	**+38.4%**
NPV	92.7%	97.9%	**+5.2%**

Disclaimer: IP-10/Cr values simulated. CRP/Cr mapped from serum. For algorithm validation only.